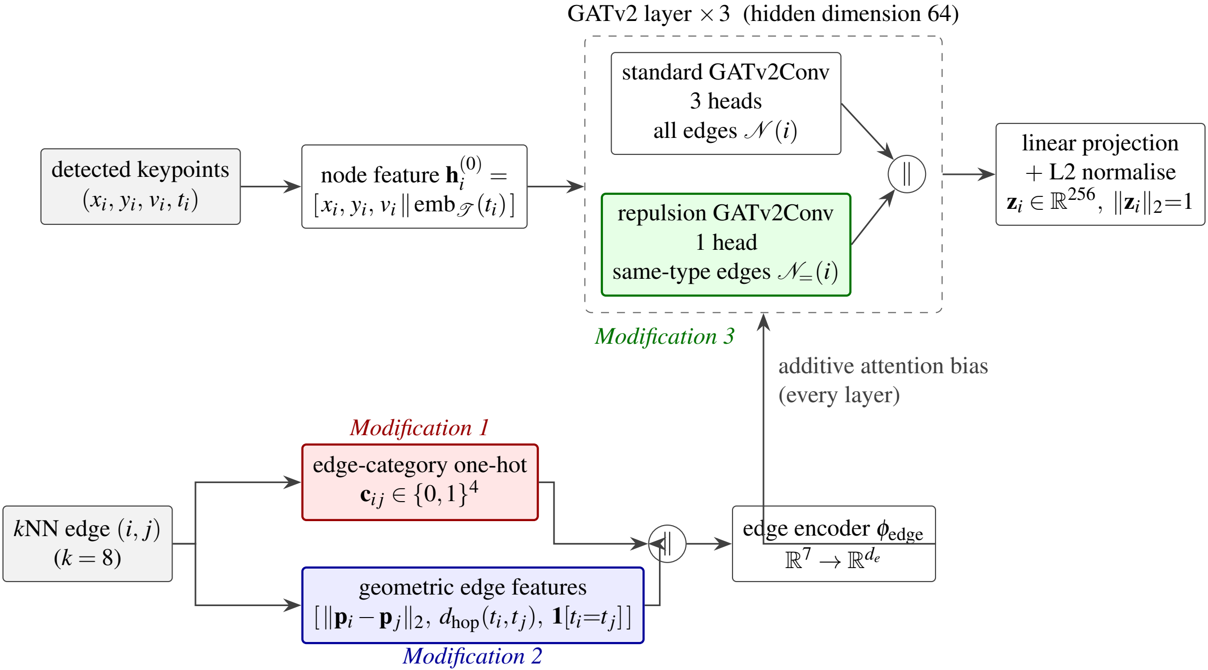

GATv2 layer × 3 (hidden dimension 64)
standard GATv2Conv
3 heads
all edges $\mathcal{N}(i)$
repulsion GATv2Conv
1 head
same-type edges $\mathcal{N}_{=}(i)$
$\|$
detected keypoints
(x_i, y_i, v_i, t_i)
node feature $\mathbf{h}_i^{(0)} =$
$[x_i, y_i, v_i \,\|\, \mathrm{emb}_{\mathcal{T}}(t_i)]$
linear projection
+ L2 normalise
$\mathbf{z}_i \in \mathbb{R}^{256}, \ \|\mathbf{z}_i\|_2 = 1$
Modification 3
additive attention bias
(every layer)
Modification 1
edge-category one-hot
$\mathbf{c}_{ij} \in \{0,1\}^4$
kNN edge (i, j)
$(k = 8)$
$\|$
edge encoder ϕ_{edge}
$\mathbb{R}^7 \to \mathbb{R}^{d_e}$
geometric edge features
$[\|\mathbf{p}_i - \mathbf{p}_j\|_2, \ d_{\text{hop}}(t_i, t_j), \ \mathbf{1}[t_i = t_j]]$
Modification 2